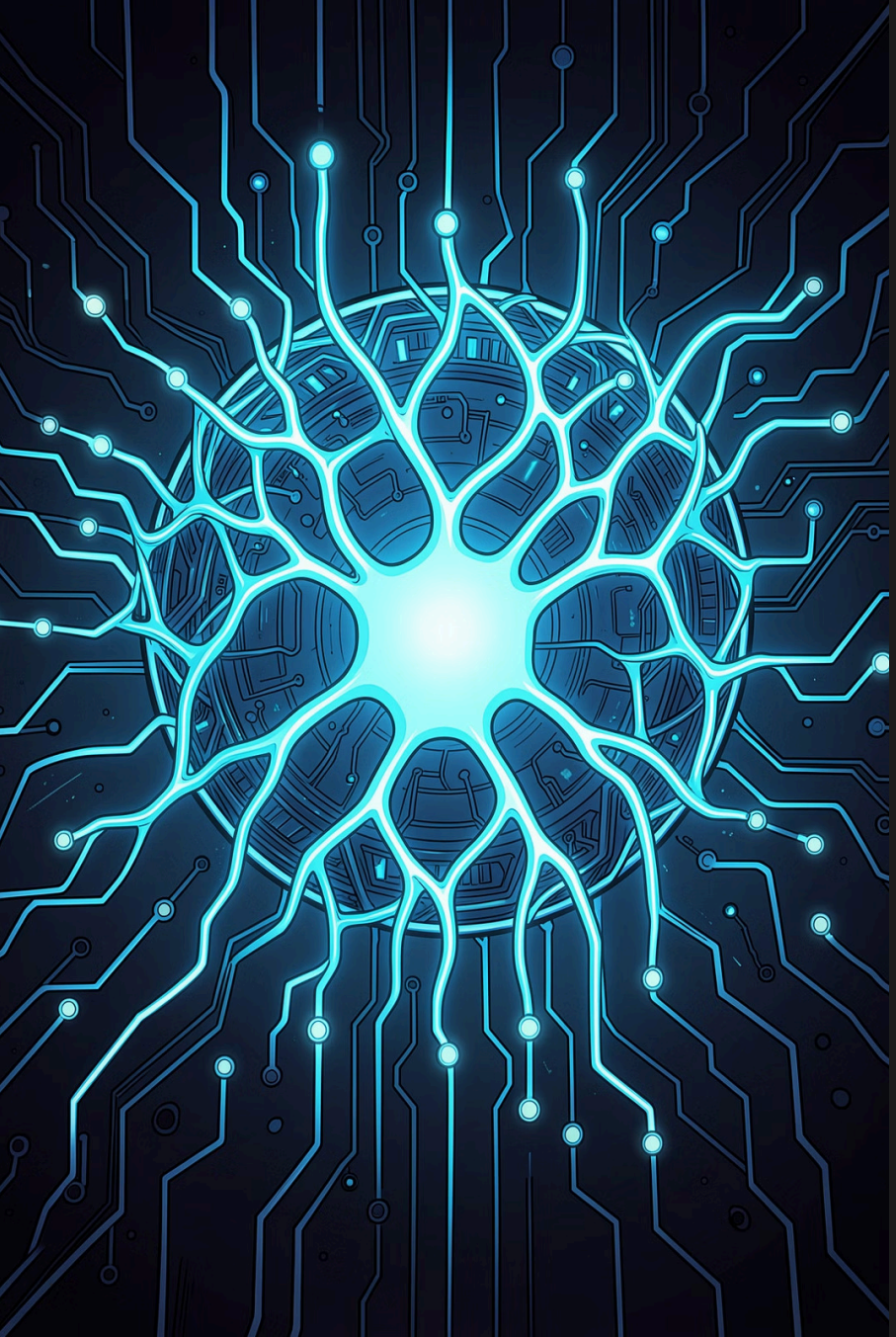


Student Performance Prediction System

Using Machine Learning

AIML SUMMER INTERNSHIP 2026

IIHMF, MNNIT ALLAHABAD, PRAYAGRAJ

Sejal Singh | United College of Engineering and Research, Prayagraj | June 2026

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Introduction

Traditional academic evaluation relies on **summative assessments** — warnings arrive only after exams conclude. This project shifts the paradigm to **proactive diagnostic intervention** using AI/ML.

The Gap

No dynamic tools interpret how sleep, social media, and attendance affect final grades before results are declared.

The Solution

An automated, secure web platform delivering real-time grade predictions and behavioural advisories.

The Impact

Enables institutions to perform diagnostic evaluations **before** summative assessment cycles close.



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Problem Statement & Objectives

Problem Statement

Academic environments lack tools to interpret how **non-scholastic parameters** — sleep deprivation, attention fragmentation, low attendance — affect final grades. Preventable credit drops occur because warnings arrive *after* exams.

Project Objectives

- Build a **Random Forest regression** engine mapping lifestyle metrics to a 0–100% performance scale
- Implement secure **SQLite3 + Werkzeug** authentication
- Deploy a **Flask** backend with Pickle serialisation
- Design a responsive UI using **Tailwind CSS + Chart.js**
- Generate **automated rule-based behavioural advisories**

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Dataset & Input Features

Six lifestyle and academic features feed the Random Forest model:

Feature	Type	Unit	Bounds	Significance
Attendance Ratio	Continuous	%	0–100	Engagement
Daily Study Target	Continuous	Hrs/Day	0–24	Independent focus
Previous Grade	Continuous	%	0–100	Academic baseline
Extracurricular	Categorical	Binary	0/1	Stress management
Daily Sleep	Continuous	Hrs/Day	0–24	Cognitive recovery
Social Media Usage	Continuous	Hrs/Day	0–24	Digital distraction

Values bounded via: $P_{\text{bounded}} = \max(0.0, \min(100.0, P_{\text{raw}}))$

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System Architecture & Technology Stack



Data flows from the Tailwind CSS input form through Flask session validation and input cleaning, into the serialised Random Forest model for prediction, and finally to the Chart.js output dashboard with automated advisory.

Backend

Flask micro-framework, SQLite3, Werkzeug hashing, Pickle serialisation

Frontend

Tailwind CSS (utility-first), Chart.js (dynamic gauge dials & bar charts)

ML Engine

Random Forest Regressor — ensemble of B decision trees with MSE-based splits

Security

Session-validated routes, hashed credentials, unique user tokens

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Machine Learning Model

Random Forest Regressor

An ensemble of **B independent decision trees** trained on bootstrapped subsets. Final prediction is the average across all trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

Optimal node splits minimise **Mean Squared Error (MSE)**, controlling over-fitting and handling non-linear feature boundaries.

6-dimensional input → continuous grade output (0–100%)

Evaluation Metrics

MAE

Mean Absolute Error — average magnitude of prediction errors

RMSE

Root Mean Squared Error — emphasises larger discrepancies

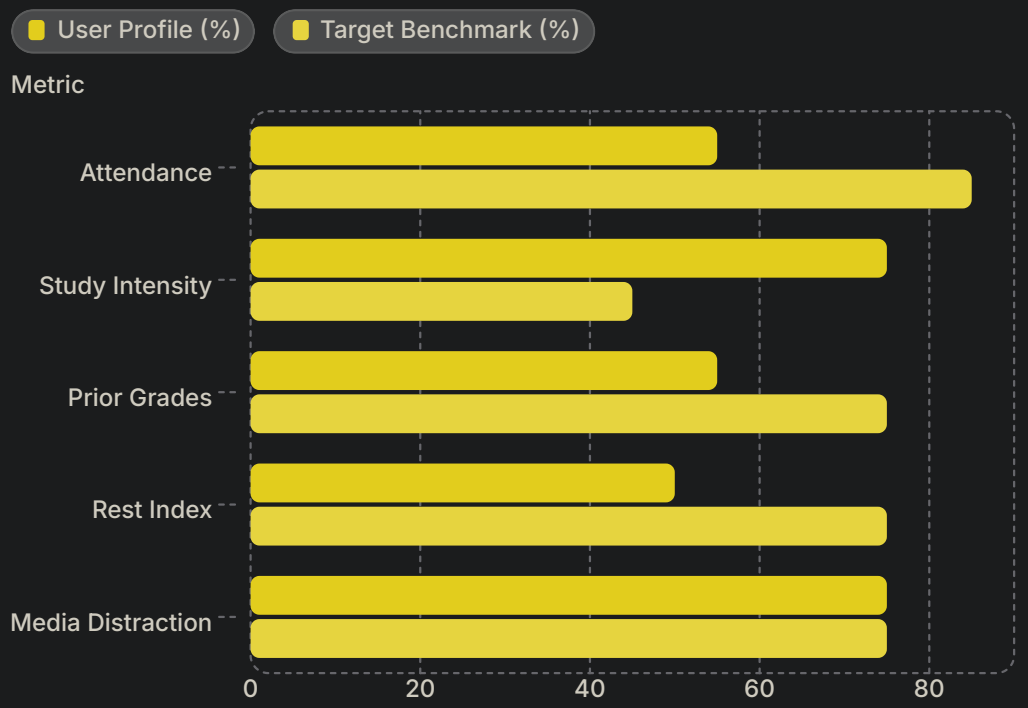
R^2 Score

Coefficient of Determination — proportion of variance explained by inputs

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Data Analysis & Visualisations

The Chart.js interface compares user metrics against benchmark targets using scaled indices:



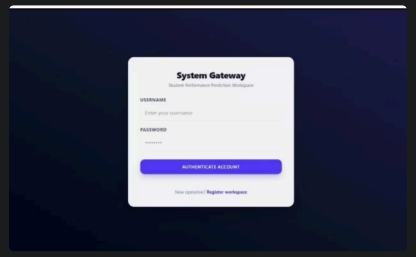
Key Observations

- **Attendance (55%)** falls well below the 85% compliance target — flagged as critical
- **Study Intensity (75)** exceeds benchmark, indicating strong independent study habits
- **Rest Index (50)** signals sleep deprivation risk — target is 7.5 hrs/day
- **Media Distraction (75)** matches benchmark but contributes to attention fragmentation

⚠ Automated advisory triggers when metrics breach safety boundaries

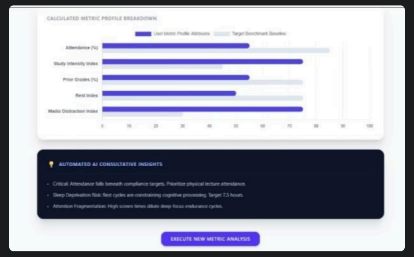
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User Interface & Screenshots



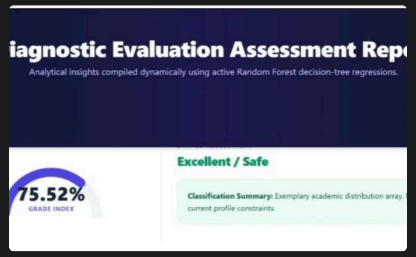
System Gateway

Secure login with username/password fields and **Authenticate Account** action button on a dark gradient background.



Input Dashboard

Structured form for all 6 lifestyle parameters — attendance, study hours, prior grades, extracurricular, sleep, and social media usage.



Results View

Displays predicted Grade Index (e.g., **75.52%**), status badge (Excellent/Safe), and automated AI consultative insights.

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Results & Performance

Sample Prediction Output

Real-time predictions are accompanied by **rule-based behavioural advisories** — e.g., attendance deficits, sleep deprivation risks, and attention fragmentation warnings.

Model Validation Framework

1

MAE

Average absolute prediction error magnitude

2

RMSE

Squared error average — penalises larger deviations

3

R^2 Score

Variance in grades explained by lifestyle features

Modelling non-academic lifestyle attributes provides a **highly accurate** predictive instrument for tracking student progress.

75.52%

Grade Index — Exemplary academic distribution

Excellent / Safe

Classification status with green badge indicator

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Conclusion & Future Scope

Achievements

Successfully deployed an end-to-end ML web platform bridging Random Forest regression with a responsive Tailwind CSS + Chart.js interface at IIHMF MNNIT.

Benefits

Shifts institutions from **reactive grading** to **automated real-time diagnostic guidance**, supporting academic retention before exams conclude.

Future Scope

Time-series trend analysis across semesters; automated background pipeline for periodic model retraining with anonymised student data.

Thank You — Sejal Singh | AIML Summer Internship 2026 | IIHMF, MNNIT Allahabad, Prayagraj

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